AURA: AUdio-dRiven streaming Avatar

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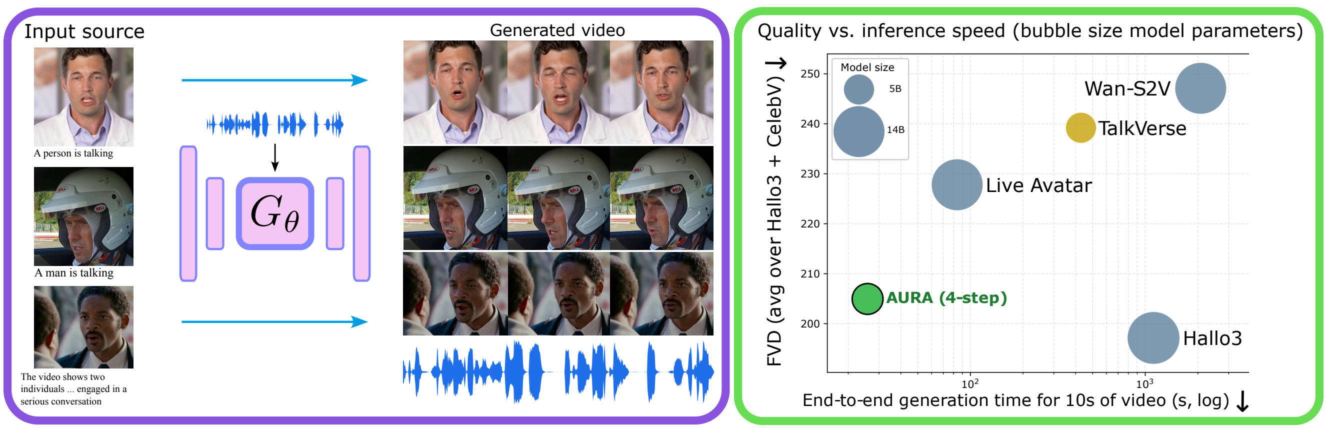

Figure 1. **AURA generates 720p talking-head videos at 9.7 fps on a single GPU in a causal streaming manner**, per chunk of 0.64s. *(Left)* Excerpts of three samples generated by our causal speech-to-video model G_θ from the same speech audio with different reference visual identities and text prompts, with frames temporally aligned across rows. Full video results are available in the supplementary material. *(Right)* Quality–inference time trade-off for AURA and baselines, averaged across the CelebV-HQ and Hallo3 test sets. We report Fréchet video distance (FVD) vs. end-to-end generation time for 10 s of video (log scale, lower is better for both); circle areas are proportional to parameter count. TalkVerse, Hallo3, and Wan-S2V are non-causal; AURA and Live Avatar are streaming.

Abstract

Current audio-driven video diffusion models can synthesize realistic talking-head videos but generally require dozens of diffusion denoising steps over multi-second video chunks, making them unsuitable for streaming applications such as voice assistants and virtual avatars, where low latency and online causal generation are required properties. AURA generates high-resolution video chunk-by-chunk at 9.7 fps with a rolling KV cache, requiring only the audio of the current chunk and thus introducing no delay between audio capture and video generation, making it well-suited to interactive avatar applications. Our two-stage distillation pipeline first aligns the causal student with the teacher, by training it to match the teacher's trajectories with fewer diffusion steps and a chunk-wise causal attention mask, then applies Distribution Matching Distillation (DMD) with backward simulation to close the remaining quality gap. On standard talking-head benchmarks, AURA matches the teacher's quality while running 17× faster, achieving 720p avatar streaming on a single GPU. To support reproducibility and further research, we plan to release our model's weights along with a dataset of 30K synthetic videos and their precomputed teacher trajectories.

1. Introduction

Recent speech-to-speech models [9, 29] enable real-time full-duplex voice interaction, paving the way for natural voice assistants. However, these models only generate audio. Human spoken communication also relies on visual cues and facial expressions: lip motion alters perceived phonemes [26], visual cues sharpen comprehension in noise [32], and an embodied talking face enhances social presence [4]. Audio-driven talking-head models address this by synthesizing video from a reference image and speech audio, enabling virtual avatars, synthetic presenters, and game characters. All such applications share a common constraint: the avatar must be generated on-the-fly, without delay, from an unbounded audio stream.

State-of-the-art talking-head models such as TalkVerse [36] and Hallo3 [6] produce high-quality results but are ill-suited to streaming: they rely on bidirectional attention and must run dozens of denoising steps over full multi-second chunks, taking minutes per chunk on a single GPU. Recent distillation methods [16, 42] compress such teachers into causal few-step students, but target only text-to-video or image-to-video settings. Extending this to audio-driven generation has so far been shown only on much larger models (14–18B parameters) [17, 24, 30], which remain too slow for real-time single-GPU inference. A further obstacle is the scarcity of suitable training data: existing public datasets consist of clips too short for distillation from teachers like TalkVerse.

We introduce *AUdio-dRiven streaming Avatar* (AURA), a 5B-parameter causal few-step speech-to-video (S2V) model that streams 720p talking-head video at 9.7 fps on a single GPU, a $17\times$ speedup over its teacher. AURA is trained via a two-stage distillation pipeline: Stage 1 aligns the causal student with the teacher's Ordinary Differential Equation (ODE) trajectories; Stage 2 closes the few-step gap with Distribution Matching Distillation (DMD) and backward simulation. Inference relies on a rolling KV cache with bounded memory.

On standard benchmarks, AURA matches its teacher on visual quality (FVD, JEDi) and audio-visual alignment while achieving the highest throughput among single-GPU audio-driven baselines. Figure 1 illustrates qualitative results and the speed–quality tradeoff against prior work.

2. Related Work

Diffusion Models and Video Generation. Image and video generation has evolved from GANs [2, 13, 18] to diffusion models [11, 15, 28, 31] and flow matching [21, 23], which model data through iterative refinement at the cost of slow sampling. Large-scale video generators such as Sora [3], HunyuanVideo [19], and Wan [35] adopt DiT architectures in compressed latent spaces, scaling to 720p–1080p over 50+ non-causal denoising steps, making them ill-suited for streaming applications.

Distillation of Diffusion Models. The slow inference of diffusion models has motivated a large body of work on *distillation* aiming to train a faster, few-step student to reproduce the output distribution of a slow teacher in far fewer steps. A particularly successful line of work is Distribution Matching Distillation (DMD) [41], which trains a one-step student to minimize $\mathrm{KL}(p_{\text{gen}} \| p_{\text{data}})$ between its distribution and the teacher's, using a learned critic to estimate the score gap. DMD2 [40] extends this framework by generalizing the student from a single denoising step to a few-step generator, and introduces a *backward simulation* strategy in which the DMD loss is computed on noisy intermediate states sampled from the student's own multi-step rollouts rather than from forward-diffused data, exposing the loss to the error trajectory the student actually follows at inference instead of the forward-diffusion path assumed by standard DMD. CausVid [42] successfully applies DMD for distilling a non-causal video diffusion teacher into a causal few-step student suitable for streaming text-to-video applications. A following line of work addresses the train/test gap via self-conditioning during training [7, 16, 22, 38, 45]. In this work, we investigate extending CausVid and DMD2 for speech-to-video talking head generation.

Audio-Driven Talking Heads. Current state-of-the-art talking-head models leverage pretrained video DiTs: Hallo3 [6] adapts CogVideoX [39] for portrait animation, OmniHuman [20] extends DiT-based generation to full-body animation, and TalkVerse [36], our teacher, builds on Wan2.2-5B to produce 720p video via 50-step non-causal denoising. All these approaches operate offline on fixed-length clips, precluding streaming use. In the streaming regime, Talking Machines [24] first addressed this setting via DMD distillation with sparse causal attention (weights unreleased). Live Avatar [17] and SoulX-FlashTalk [30] distill 14B-parameter teachers into few-step students but require 4–5 GPUs for real-time throughput at 480p. Closest to our work, Avatar Forcing [8], whose pre-print was concurrently published, uses a similar DMD-based pipeline but requires a look-ahead window of future audio by design, introducing an incompressible streaming delay. In contrast, AURA distills a 5B teacher into a causal 4-step student delivering 720p at 9.7 fps on a single GPU with no audio look-ahead.

3. Method

AURA distills a non-causal audio-driven video diffusion teacher (TalkVerse) into a 4-step causal student, streaming 720p avatars in constant memory via a two-stage pipeline illustrated in Fig. 2.

Preliminaries: DMD and Backward Simulation. Distribution Matching Distillation (DMD) [41] trains a few-step student G_θ to match the teacher at the distribution level by minimizing, in expectation over the diffusion timestep $t \sim \mathcal{U}(0, 1)$, the KL divergence $\mathrm{KL}(p_{\text{gen},t} \| p_{\text{data},t})$ between the student distribution and the data distribution, both diffused to noise level t. While these densities are intractable, the gradient admits a closed-form expression as a score gap evaluated on student samples:

$$\nabla_\theta \mathcal{L}_{\text{DMD}} \approx -\mathbb{E}_t \int \big(s_{\text{data}}(\hat{x}_t, t) - s_{\text{gen},\eta}(\hat{x}_t, t)\big) \tfrac{dG_\theta(z)}{d\theta}\, dz, \tag{1}$$

where $\hat{x}_t = \Psi(G_\theta(z), t)$ is the forward-diffused student sample, s_{data} is the frozen teacher score, and $s_{\text{gen},\eta}$ is a trainable critic updated jointly with G_θ. DMD2 [40] extends this with *backward simulation*: rather than computing the DMD loss on forward-diffused real samples, the student is first unrolled for a few denoising steps to produce synthetic intermediate states $\hat{x}_t$, aligning the training distribution with the one the student actually encounters at inference.

AURA Architecture. TalkVerse [36] is a 5B-parameter DiT built on Wan2.2-5B, conditioned on a reference image, Wav2Vec2-encoded speech [1] injected via cross-attention, and a T5-XXL text prompt [27]. Its VAE uses (t, h, w)=$(4, 16, 16)$ compression, enabling efficient 720p generation. We introduce three architectural modifications to turn this 50-step bidirectional teacher into a 4-step causal student that is suited to real-time streaming.

Chunked causal attention. The student generates video in chunks of $n_c = 4$ latent frames ($\sim$0.64 s). A block-wise causal mask prevents chunk C_i from attending to future chunks C_j, $j>i$, while attention remains bidirectional within each chunk.

No motion frames. TalkVerse conditions each window on past motion frames for temporal coherence; our KV cache already carries this context, so we drop them.

Per-chunk mixed timesteps. We train the student to handle sequences in which different chunks sit at different noise levels, assigning a single diffusion timestep t_i per chunk. This is what later enables a compact KV cache at inference (see the rolling KV cache paragraph below).

Stage 1 — ODE Teacher Forcing. Directly applying DMD to a causal student is unstable when the student architecture differs from the teacher [42]. Stage 1 therefore initializes the causal student by regressing it onto teacher ODE trajectories. Our training dataset provides latents at five timesteps $\{0, 0.25, 0.5, 0.75, 1\}$ along the teacher's 50-step ODE. At each step, the student is queried at a per-chunk mixed-timestep configuration sampled uniformly from $\{0.25, 0.5, 0.75, 1\}$ and trained to predict the clean latent x_0 via MSE:

$$\mathcal{L}_{\text{stage1}} = \mathbb{E}_{t, z_t, c}\big[\|\hat{x}_0(z_t, t, c) - x_0\|_2^2\big]. \tag{2}$$

Stage 2 — Backward Simulation DMD. Stage 2 applies DMD with backward simulation to close the few-step quality gap. We maintain three networks: the frozen teacher f_{teacher}, the trainable student G_θ (initialized from Stage 1), and a trainable critic f_ϕ initialized from the teacher and updated 5 times per student step. At each iteration: (i) G_θ is unrolled without gradients to obtain intermediate states $\{\hat{x}_{t_k}\}$ along its own trajectory; (ii) these states are re-fed through G_θ with gradients under per-chunk mixed timesteps; (iii) the DMD gradient is evaluated at the resulting $\hat{x}_0$:

$$\nabla_\theta \mathcal{L}_{\text{DMD}} = \mathbb{E}_{t,\,\hat{x}_0}\left[\big(f_\phi(\hat{x}_t, t) - f_{\text{teacher}}(\hat{x}_t, t)\big)\frac{\partial \hat{x}_0}{\partial \theta}\right], \tag{3}$$

where $\hat{x}_0 = G_\theta(z, c)$, z is the noise input, c denotes the conditioning (audio, prompt, and reference image), and $\hat{x}_t = \Psi(\hat{x}_0, t)$. This exposes the student to the states it will encounter at inference.

Inference — Rolling KV Cache. At inference, the student generates one chunk at a time with 4 denoising steps and maintains a KV cache of fixed size. A naive cache grows along two independent axes: with the number of denoising steps (one entry per step) and with video length (one entry per past chunk). Our design removes both. First, mixed-timestep training lets each past chunk be cached at a single noise level, its final, lowest-noise state, rather than one entry per denoising step, so a single cache is shared across all 4 steps (4$\times$ reduction). Second, a *rolling* window keeps only two chunks: the **first** as a fixed anchor (an attention sink [37]) for long-term identity, and the **most recent** for immediate temporal context; all others are discarded. Together these yield constant memory regardless of video duration. See Fig. 3 of the supplementary.

Training Dataset. Existing datasets are ill-suited to our setup: the TalkVerse training set is hard to obtain reliably, while Hallo3 clips are too short to cover the teacher's generation window of $\sim$5 s. We therefore build a synthetic dataset by running the TalkVerse teacher on controlled conditioning inputs: reference images and text prompts from Hallo3 [6], paired with speech synthesized from English transcripts via an off-the-shelf TTS model [43]. For each of the 30K resulting clips, we store the latent ODE trajectory at five timesteps $(0, 0.25, 0.5, 0.75, 1)$ to support Stage 1. Stage 2 requires no additional data: backward simulation generates training samples from the student's own rollouts, conditioned on the same Hallo3 images and synthetic audio tracks. **No real video data is needed at any stage**.

4. Experiments

4.1. Experimental Setup

Architecture and training. The student is initialized from TalkVerse (5B parameters) and fine-tuned with LoRA adapters of rank 128 on attention layers, plus unfrozen timestep embeddings and output head. Videos are generated at 720p in chunks of $n_c = 4$ latent frames with 4 denoising steps per chunk. Training proceeds in two stages: Stage 1 (ODE teacher forcing, 120K steps) aligns the causal student with the teacher's trajectories; the Stage 1 LoRA is then merged and re-initialized for Stage 2 (DMD with backward simulation, 5K steps), which closes the few-step quality gap. All experiments run on 64 H100 GPUs.

Baselines. We compare against *TalkVerse* [36] (our teacher 5B), *Wan-S2V* [12] and *Hallo3* [6] (non-causal baselines $\sim$14B)), and *Live Avatar* [17] (our closest competitor, streaming few-step, 14B). All baselines use released checkpoints under recommended settings.

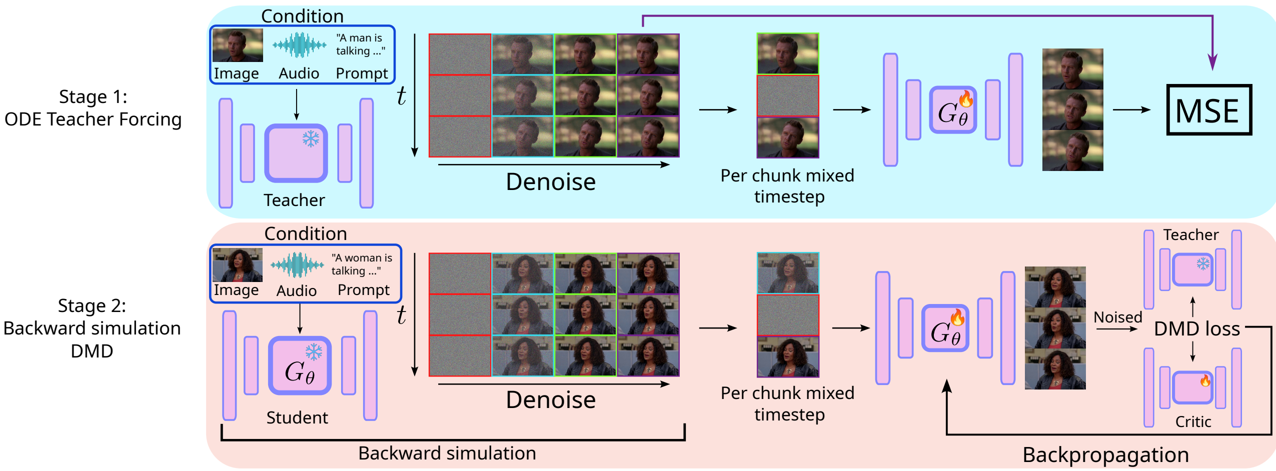

Figure 2. **Two-stage distillation pipeline**. *Stage 1 (top, ODE teacher forcing)*: the frozen teacher generates a multi-step ODE trajectory; a per-chunk mixed-timestep state is sampled and fed to the causal student G_θ, trained to predict clean latents under MSE. *Stage 2 (bottom, DMD with backward simulation)*: the student generates a video without gradients (backward simulation), and the DMD loss is computed by comparing teacher and critic scores on states from the student's own trajectory.

Table 1. **Main results** on HALLO3 TEST (H) and CELEBV-HQ TEST (C). FVD and JEDi are averaged over three temporal offsets. TalkVerse and Hallo3 use 50 denoising steps, Wan-S2V 40, and Live Avatar and AURA 4; TalkVerse and AURA are 5B, other baselines ∼14B. WE: warping error. **Bold**: best; underlined: second best.

Method	Test	fps↑	FID↓	FVD↓	JEDi↓	CSIM↑	Sync-C↑	Sync-D↓	WE↓
Hallo3	H	0.22	**43.1**	186.3	**0.16**	0.74	5.06	9.9	0.0036
Wan-S2V	H	0.08	55.2	200.5	0.27	0.71	**5.40**	**9.0**	0.0032
TalkVerse	H	0.58	55.7	215.5	0.42	0.74	3.82	10.5	**0.0019**
Live Avatar	H	1.9	50.8	**182.7**	0.47	0.77	4.90	9.3	0.0039
AURA	H	**9.7**	57.2	191.0	0.37	**0.80**	4.09	10.0	0.0034
Hallo3	C	0.22	**36.0**	**208.0**	**0.257**	0.71	**6.20**	9.4	**0.0038**
Wan-S2V	C	0.08	46.7	293.7	0.640	0.62	**6.20**	**8.9**	0.0061
TalkVerse	C	0.58	56.3	262.9	0.600	0.59	4.80	10.5	0.0059
Live Avatar	C	1.9	43.1	272.9	1.087	0.67	5.90	9.2	0.0057
AURA	C	**9.7**	45.9	218.9	0.477	**0.74**	4.70	10.1	0.0055

Benchmarks and metrics. We evaluate on two test sets. HALLO3 TEST (100 clips, 10 s) pairs first frames from the Hallo3 test set with real English speech; CELEBV-HQ TEST (150 clips, 8–20 s) pairs CelebV-HQ [44] frames with TTS-synthesized speech. We report FID [14], FVD [34] and JEDi [25] (an MMD-based metric on V-JEPA features, more reliable at our sample counts) for visual quality, Sync-C/D [5] for lip-sync, CSIM (ArcFace [10] cosine similarity) for identity preservation, and warping error (WE) via RAFT optical flow [33] for temporal consistency. All timings are measured on a single H100 GPU.

4.2. Results

Table 1 reports our main quantitative results. AURA outperforms its TalkVerse teacher across most quality metrics despite using only 4 denoising steps instead of 50, and is on par with the much larger Wan-S2V (14B) on FVD, JEDi, and audio-visual alignment. On identity preservation (CSIM), AURA outperforms all baselines on both test sets (0.80 / 0.74), which we attribute to the rolling cache strategy that retains the first generated chunk as a long-term identity anchor. AURA is also the fastest method: 9.7 fps on a single H100, a 17× speedup over TalkVerse and 5× over Live Avatar. Figure 1 (right) summarizes the quality–speed trade-off: AURA matches the best baselines on quality while being one to two orders of magnitude faster. While Hallo3 achieves the best FID and FVD, visual inspection reveals exaggerated mouth motion and hand artifacts that distribution-based metrics fail to penalize. An **user study** (supplementary, Sec. 7) confirms this: raters prefer AURA over Hallo3 on all four axes, and rank it on par with its (much slower) TalkVerse teacher on lip-sync and naturalness. Qualitative examples are shown in Figure 1 (left). Video results are provided in the supplementary material.

5. Conclusion

We present AURA, a 5B causal few-step speech-to-video model that distills a bidirectional teacher into a 4-step causal student via ODE-trajectory initialization and DMD with backward simulation, with streaming inference through a fixed-size rolling KV cache. On standard talking-head benchmarks, AURA matches its teacher on FVD and JEDi while streaming 720p video at 9.7 fps, a 17× speedup. Occasional chunk-boundary discontinuities and the gap to the 25 Hz rate needed for real-time single-GPU generation are natural directions for future work.

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AURA: AUdio-dRiven streaming Avatar

Supplementary Material

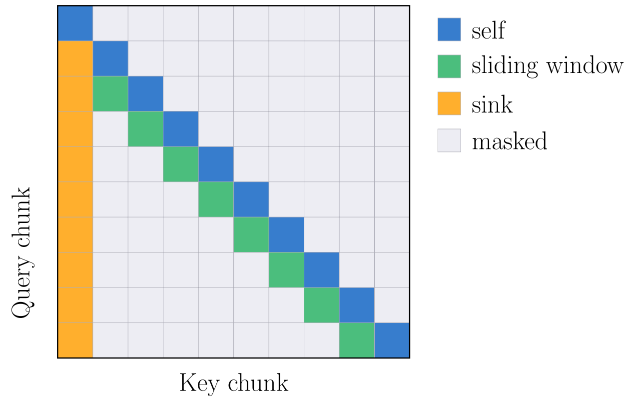

Figure 3. **Sliding-window causal attention mask (training).** Each chunk of 4 latent frames attends to itself, the preceding chunk, and the initial chunk acting as an attention sink. Attention is bidirectional among the 4 frames within a chunk.

6. KV cache figure

Figures 3 and 4 detail the attention and caching mechanisms underlying constant-memory inference. During training (Fig. 3), a sliding-window causal mask lets each chunk attend only to itself, the immediately preceding chunk, and the initial chunk, which acts as a persistent attention sink for long-term identity.

A naive KV cache grows along two independent axes: the number of denoising steps and the video length. Figure 4 shows how per-chunk mixed-timestep training removes the first axis. Without it (left), a distinct cache must be kept for every denoising step, since each step operates at a different noise level; memory then scales linearly with the number of steps T. With mixed-timestep training (right), the student is trained to process chunks held at different noise levels simultaneously, so each past chunk can be cached once, at its final low-noise state, and reused across all T steps. Combined with the rolling window over chunks, which bounds the second axis, this yields memory that is constant in both the number of steps and the video duration.

7. User Study

We run an anonymous human evaluation on the Mabyduck platform (30 raters, 300 pairwise comparisons) on the 100 Hallo3 test clips, comparing AURA, Hallo3 [6], and TalkVerse [36] for overall quality, lip-sync, reference fidelity and naturalness (Fig. 5). AURA is preferred over Hallo3 on all axes, confirming that Hallo3, despite strong

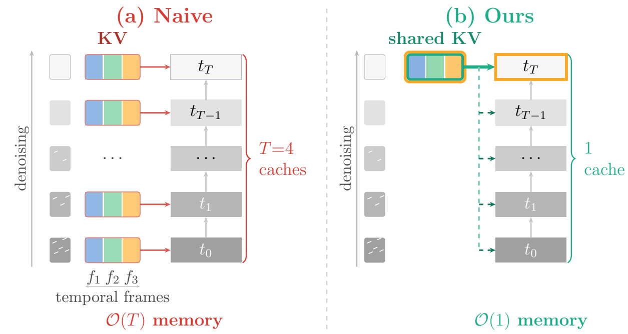

Figure 4. **KV caching with and without per-chunk mixed timesteps.** *Left:* a naive scheme stores one cache per denoising step, so memory scales linearly with the number of steps T. *Right:* with mixed-timestep training, each past chunk is cached at a single noise level, sharing one cache across all steps.

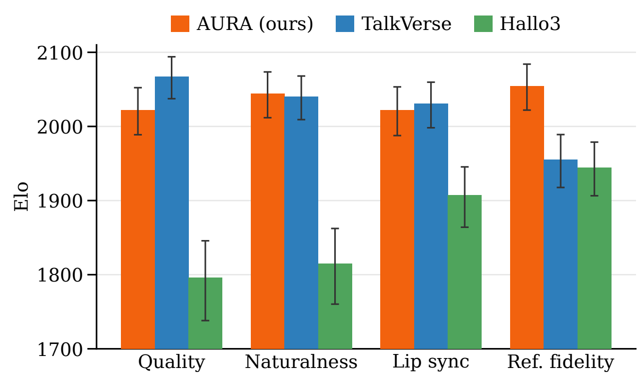

Figure 5. **Human preference across four axes.** Bayesian Elo (↑) from pairwise comparisons; error bars are 95% CIs.

benchmark results, suffers from visual artifacts and reduced naturalness. Against the non-causal teacher TalkVerse, AURA ranks higher on reference fidelity (in line with our CSIM results), on par for lip-sync and naturalness, and only marginally below on visual quality. This confirms that AURA's lip-sync is not degraded and that the perceptual gap with TalkVerse is small.